IMPERIAL

Sonar-Based Underwater SLAM with 3D Gaussian Splatting Mapping

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Why Underwater Sonar SLAM Is Hard

Can a renderable sonar map provide online correction factors?

1. Problem setting

Underwater robots cannot use GNSS. IMU/DVL odometry therefore drifts over time.

2. Sensor limitation

Sonar works underwater, but the image evidence is sparse, ambiguous, and viewpoint-dependent.

3. Classical SLAM limitation

Geometric matching becomes fragile under sonar noise and repeated acoustic structure.

4. Key insight from 3DGS

A learned Gaussian scene can predict the sonar view from a candidate pose.

5. Gap

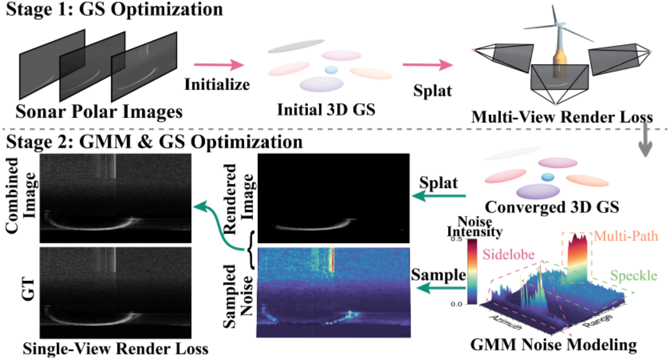
3DGS is rarely used as an online sonar measurement model inside SLAM.

6. Project idea

Use render consistency between live sonar and NAS-GS prediction as a Pose3 SLAM constraint.

NAS-GS: Noise-Aware Sonar Gaussian Splatting

A renderable sonar measurement model reference



NAS-GS training pipeline, adapted from Xu et al.

What NAS-GS does

- Learns a 3D Gaussian sonar scene from polar sonar images.
- Optimises rendered structure together with acoustic noise appearance.
- Produces a differentiable sonar renderer for new viewpoints.

How I use it

- Freeze the trained map during replay.
- Render candidate Pose3 hypotheses online.
- Insert a factor only when live sonar and render agree.

Render Consistency As A Pose3 Factor

NAS-GS predicts observations; iSAM2 consumes accepted constraints

Online loop: render from the map, compare with the live frame, then add a graph factor only when the evidence is strong.

Renderable sonar map

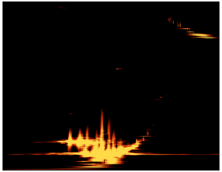
NAS-GS predicts what the sonar should observe from a proposed Pose3 camera pose.

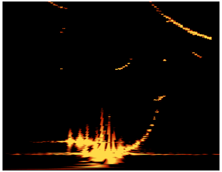
Online evidence

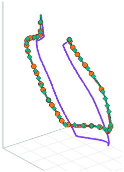
A live sonar frame is compared against rendered candidates around the current drifted pose.

Graph correction

If render similarity and gates pass, the recovered pose is inserted as a Pose3 factor.





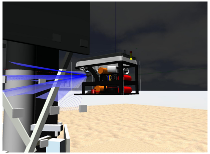


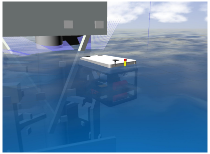
Simulation Platform: DAVE In Docker

Repeatable sonar, IMU, DVL, and ground-truth recording

Docker setting and robot model

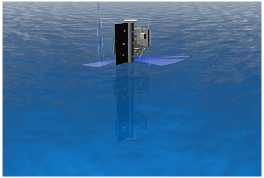
- CUDA-enabled Docker container.
- DAVE with ROS Noetic and Gazebo.
- BlueROV2 vehicle model.
- Sensors recorded: sonar, IMU, DVL, odometry, and ground truth.





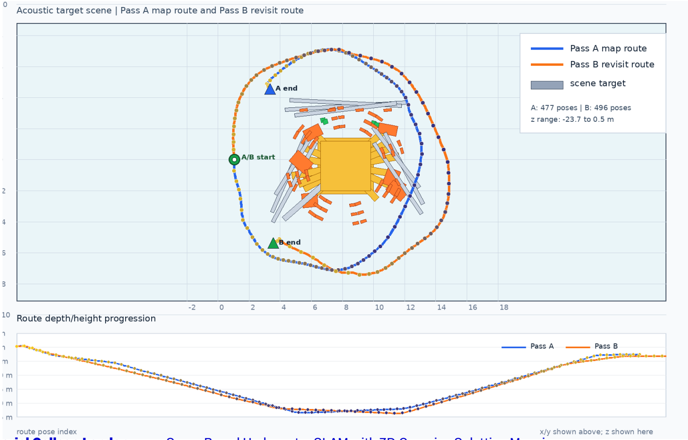
Custom scene description

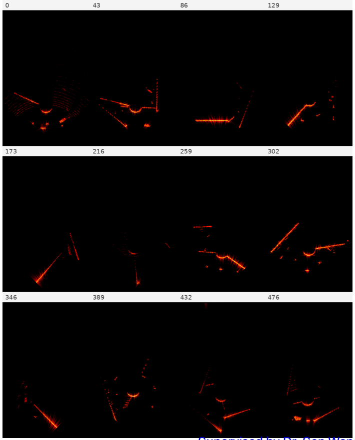
- A fixed underwater target structure creates repeatable sonar returns.
- Pass A, Pass A revisit, and Pass B use controlled waypoint routes.
- Sonar field of view, range, beam pattern, and noise are explicitly set.



Pass A Map Recording And Revisit Path

Pass A trains the NAS-GS map; Pass B reuses the start point but follows a harder revisit path

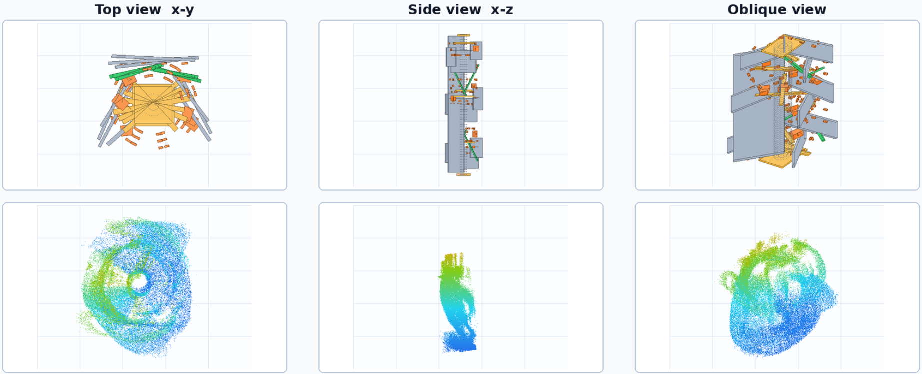




From Simulated Geometry To NAS-GS Map

The learned map is sonar-visible structure, not a full CAD reconstruction

The two rows use the same top, side, and oblique projections. They differ because the DAVE row shows the full simulated target geometry, while the NAS-GS row shows Gaussian centres learned only from sonar-visible acoustic returns.



Top row: DAVE target geometry from SDF. Bottom row: NAS-GS map centres from point_cloud.ply. Same projection and bounds per column.

Online-render Registration Backbone

Same online loop for same-route correction and revisit correction

Live replay sonar + Pose3 → Candidate poses → NAS-GS render → Score and gate → Pose3 factor → iSAM2 update

Local single

Compact x, y, z, ψ search around the live Pose3 baseline. Main same-route policy for Pass A and off-shore replay.

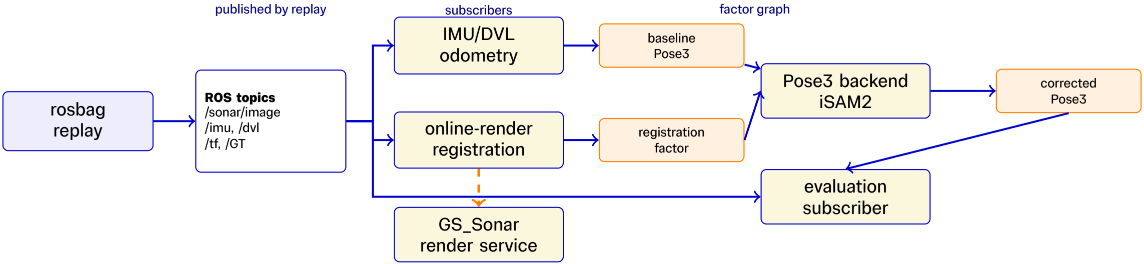
Route-progress revisit

Sequence render search guided by route progress and drift-prior checks. Used for revisit cases such as Pass B.

Gating is separate from the mode. Relaxed gating accepts useful factors earlier, while score and structure checks still prevent weak render matches from entering the graph.

Algorithm Step 1: Live Replay System

Topics are replayed, subscribed, rendered, and fused as Pose3 factors

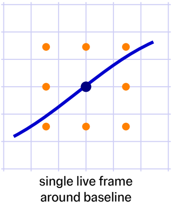


The replay is deliberately online. Nodes subscribe to ROS topics in time order, the registration node queries NAS-GS only at candidate poses, and iSAM2 receives only gated Pose3 factors.

Algorithm Step 2: Candidate Pose And NAS-GS Render

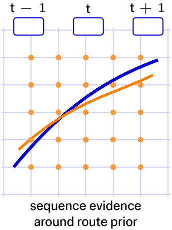
The system searches poses, not image transformations

Local single-frame search



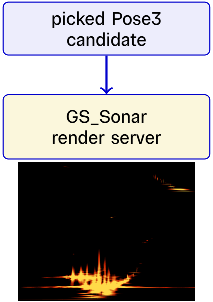
Compact x, y, z, ψ grid for same-route correction.

Route-progress sequence search



The orange line is a smooth candidate correction sequence. Wider candidates are allowed, but neighbouring frames must agree.

Render query

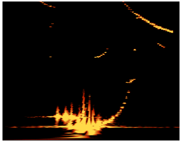


The output is a rendered sonar image for the proposed robot pose.

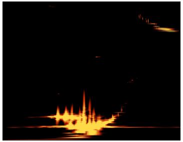
Algorithm Step 3: Score And Gate

A factor is accepted only when image evidence and pose checks agree

Accepted pair



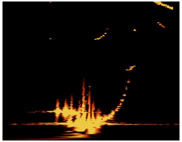
live sonar



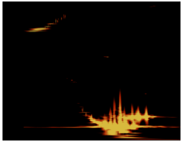
rendered sonar

Strong return overlap and low residual image difference.

Rejected pair



live sonar



rendered sonar

Visible structures do not agree, so no graph factor is created.

Score calculation

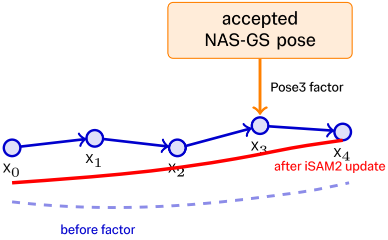
The replay normalises intensity, measures NCC-style similarity, penalises residual error, and checks bright-return overlap.

Gates

A candidate must pass render-score, structure-support, pose-jump, and sequence-consistency gates before insertion.

Algorithm Step 4: Pose3 Factor And iSAM2 Update

A passed render match becomes a graph constraint



Before

The graph contains IMU/DVL odometry factors, so drift accumulates along the chain.

Insertion

The accepted candidate pose and covariance are published as a Pose3 registration factor.

After

iSAM2 optimises the graph, pulling nearby states toward the render-consistent pose.

Algorithm Step 5: Route-progress Revisit Priors

Pass B uses sequence evidence plus route and drift priors

Pass B window → route progress selects map window → drift prior sets envelope → sequence render search → score and gate → Pose3 factor

Route prior

$$s_t = \frac{t}{N_B - 1}, \qquad i_t = \mathrm{round}(s_t(N_A - 1))$$

- t is the current Pass B frame index.
- N_A, N_B are route lengths.
- s_t is normalised progress.
- i_t selects the Pass A map window.
- This avoids nearest-pose lookup when route direction changes.

Drift prior

$$T_{B,t}^{\mathrm{cand}} = T_{B,t}^{\mathrm{odom}} \oplus c_t, \qquad c_t \in \mathcal{E}_{\mathrm{drift}}(s_t)$$

- $T_{B,t}^{\mathrm{odom}}$ is the live drifted Pose3.
- c_t samples correction in x, y, z, ψ.
- $\mathcal{E}_{\mathrm{drift}}$ is the progress-dependent envelope.
- It is learned from Pass A baseline-to-GT errors.
- It guides search size, not the final accepted pose.

Experimental Design

Recorded data, map training, and replay policy

Case	Images	Sonar setting	NAS-GS map	Replay policy
Pass A	477	90°, 20 m	Pass A, 900k/i1000	Relaxed local single
Pass A revisit	431	90°, 20 m	Pass A, 900k/i1000	Relaxed local; route-progress check
Pass B revisit	496	90°, 20 m	Pass A, 900k/i1000	Relaxed route-progress sequence
Offshore	1139	120°, 12 m	Offshore, 750k/i800	Relaxed local single

Baseline setting

IMU/DVL Pose3 replay with NAS-GS factors disabled. Drift is injected as DVL scale error, velocity bias, yaw drift, dropout, and replay geometry, while sonar map frames stay clean.

Relaxed gating

Minimum render score 0.08, maximum accepted correction 5 m in x, y and 8 m in z, followed by render-validity and factor-consistency checks.

Pass A Policy Selection

Rendered BP library with runtime no-BP was the most reliable local policy

Pass A online-render with BP and no-BP rendering

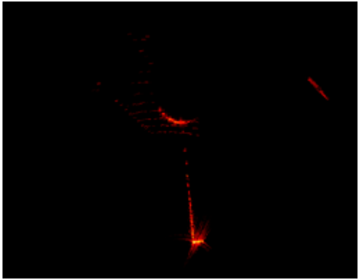
Same relaxed local single policy with different NAS-GS render settings

Pass A Sonar Evidence

One recorded frame and the corresponding BP/no-BP rendered libraries

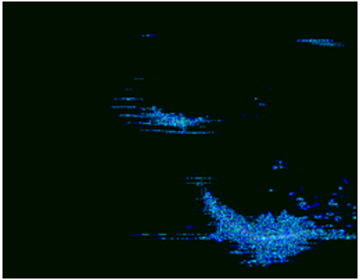
Recorded Pass A

Live sonar frame from the DAVE route.



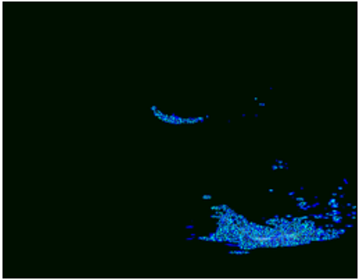
Rendered BP library

Reference image from the trained NAS-GS map.



Rendered no-BP library

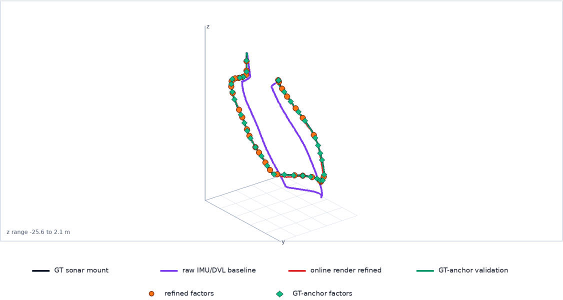
Alternative rendered appearance for policy testing.



Pass A Result

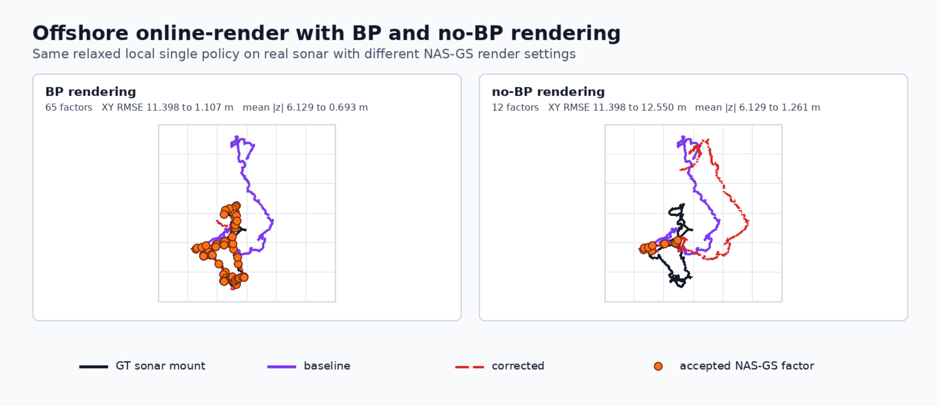
The same-route case shows the strongest online-render correction with the selected policy

Pass A 3D z-drift diagnostic with map-pose registration



BP versus no-BP on Offshore Data

The better rendering mode depends on the recorded sonar appearance



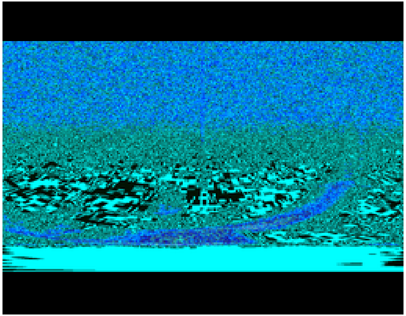
Offshore data contains stronger real-image artefacts, so BP and no-BP rendering are less interchangeable than in the DAVE Pass A replay.

Offshore Sonar Evidence

The real replay has stronger appearance mismatch than the DAVE route

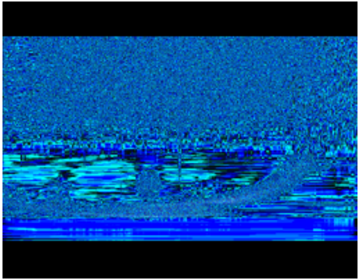
Recorded offshore

Live sonar from the replay.



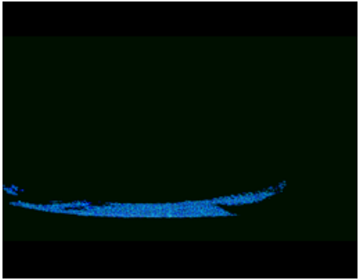
Rendered BP library

NAS-GS render with the offshore BP library.



Rendered no-BP library

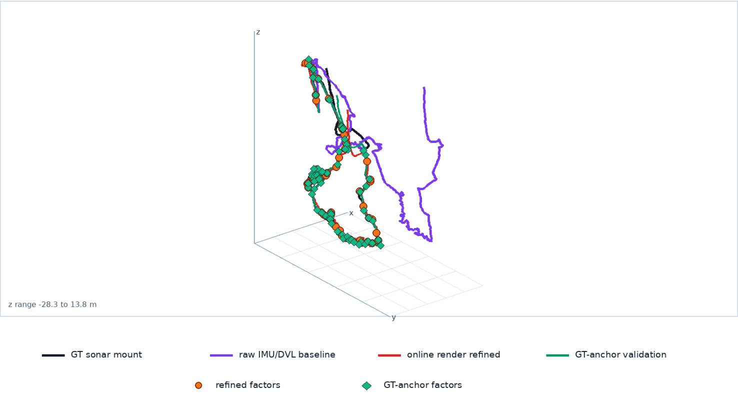
No-BP render used for appearance comparison.



Offshore Validation

The method transfers, but the real-scene case exposes z and appearance limits

Offshore 3D pose graph with live online-render correction



GT-anchor curve is the archived upper-bound diagnostic. Baseline and online-render are current live replay results.

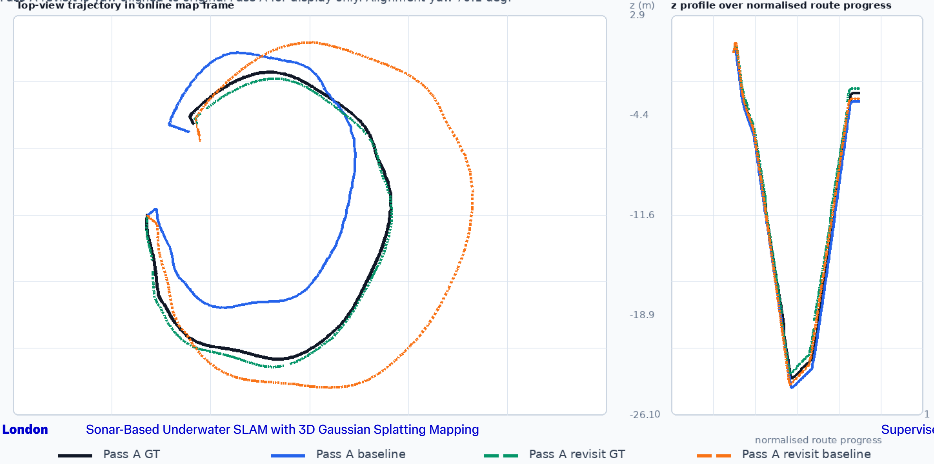
Revisit Baselines

Pass A revisit is same-route; Pass B is harder because the route direction changes

Pass A and Pass A revisit realistic-drift baselines

Top-view uses the graph baseline. The z panel uses raw IMU/DVL odometry, where baseline spans 24.7 m and 24.8 m against GT spans 23.9 m and 23.9 m.

Pass A revisit is yaw-aligned to original Pass A for display only. Alignment yaw 70.1 deg.



Revisit Result

The revisit policy improves the hard case conservatively

Route-progress and same-route revisit replay results

Local single is strong for the same-route Pass A revisit. Route-progress gives guarded partial correction for the harder revisit cases.

What The Results Show

The method works best as selective map-aided correction

Evidence

- Pass A shows strong same-route correction from a trained NAS-GS map.
- Pass A revisit confirms the local policy transfers to a newly recorded repeat route.
- Offshore BP replay transfers the same backbone to real sonar appearance.
- Accepted factors are interpretable because each one has a rendered-image explanation.

Boundary

- This is not continuous sonar odometry or global relocalisation.
- Reverse-route revisit improves on average, but endpoint stability is not solved.
- Wide search needs sequence confirmation before large corrections are trusted.
- Offshore accuracy is reference-consistency against Aqua-SLAM, not independent survey truth.

The useful operating region is repeated inspection: a well anchored map supplies selective Pose3 factors that correct drift when the live sonar remains inside a recoverable search basin.

Future Work

The next steps follow directly from the remaining failure modes

Algorithm

- Strengthen sequence evidence for large-drift recovery.
- Improve route-progress endpoint stability for reverse revisits.
- Use drift priors as guarded envelopes, not direct pose predictions.

Validation

- Retrain or fine-tune NAS-GS with corrected trajectories.
- Repeat offshore tests with an independent reference if available.
- Move from replay to real robot timing, calibration, and safety tests.

Conclusion

NAS-GS can act as a useful sonar map prior for SLAM.

The implemented system turns render consistency into selective Pose3 registration factors and fuses them with IMU/DVL odometry through iSAM2.

It is strongest for same-route and repeated-inspection correction. Route-progress revisit and offshore replay show the method transfers, but large-drift recovery and independent real-world validation remain the main next steps.

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Thank you

Questions?